MILITARY SPECIFICATION

TEST HARNESS, AUTOMATIC PILOT SYSTEM AN/ASM-50(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for shop use and shall be used in testing the Flight Control Group AN/ASA-32.

1.2 Classification - The Test Harness, Automatic Pilot System AN/ASM-50(*) shall consist of the following items:

Items	Type Designation	Appl. Para.
Test Set, Automatic Pilot Flight Controller	TS-1592(*)/ASM-50	3.4.1
Test Set, Automatic Pilot	TS-1593(*)/ASM-50	3.4.2
Rack, Amplifier	MT-2595(*)/ASM-50	3.4.3
Bracket, Limiting Accelerometer	MT-2596(*)/ASM-50	3.4.4
Bracket, Lateral Accelerometer	MT-2597(*)/ASM-50	3.4.5
Mount, Rate Gyroscope	MT-2598(*)/ASM-50	3.4.6
Cable Assy., Special Purpose, Electrical	W-1 of AN/ASM-50	3.4.7
Cable Assy., Special Purpose, Electrical	W-2 of AN/ASM-50	3.4.8
Cable Assy., Special Purpose, Electrical	W-3 of AN/ASM-50	3.4.9
Cable Assy., Special Purpose, Electrical	W-4 of AN/ASM-50	3.4.10
Cable Assy., Special Purpose, Electrical	W-5 of AN/ASM-50	3.4.11
Cable Assy., Special Purpose, Electrical	W-6 of AN/ASM-50	3.4.12
Cable Assy., Special Purpose, Electrical	W-7 of AN/ASM-50	3.4.13

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

FSC 4940

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SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-H-15362	Harnesses; Test Bench, Avionic Equipment, General Specification For
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Specification For
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for

STANDARDS

Military

MIL-STD-242	Electronic Equipment Parts

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

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(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.1.2 Electron Devices - Transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The equipment shall fulfill all design, workmanship and construction requirements of Specifications MIL-H-15362 and MIL-T-21200 unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 210 pounds.

3.2.3 Operating Life

3.2.3.1 Operating Stability - The equipment shall operate with optimum performance for 100 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during normal use.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 Mean Operating Time Between Failures - The equipment shall have a minimum of 500 hours of mean operating time between failures when tested and accepted as outlined under the requirements of 4.4.2.

3.2.3.4 Totalizing Time Meter - Test Sets TS-1592/ASM-50 and TS-1593/ASM-50 shall include a time totalizing meter conforming to the applicable requirements of Specification MIL-M-7793.

3.2.4 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of Specification MIL-T-21200.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

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3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ±10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC, 3 phase, Y connected neutral ground and 27.5 ± 0.5 V DC

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 3 equipment, except as modified herein.

3.2.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-T-21200 for Class 3 equipment, and shall give specified performance from the following power source with characteristics as defined in Specification MIL-T-21200 having limits as specified herein. The power required shall not exceed the specified amounts.

(1) AC Power (Three Phase), 115V ±10%, 320 VA, 50 to 70 cps and 380 to 420 cps

(2) DC Power, 27 ±2V, 4.35 Amps

3.2.11.1 Voltage Protection - The equipment shall not be damaged by voltages either above or below the voltages specified herein and shall automatically resume operation when the voltage returns within limits.

3.2.12 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-T-21200.

3.2.13 Input Power Connections - Input power connections shall be in accordance with Specification MIL-T-21200.

3.2.14 Form Factor - The units composing the AN/ASM-50 shall be mounted on an Electric Test and Maintenance Table with overall dimensions of the combined units not to exceed 50 inches high, 44 inches wide and 42 inches long.

3.3 Performance

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Initial Conditions - There shall be no initial conditions necessary, other than connection to the test article, before operation of the equipment.

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3.3.3 Control Settings - There shall be no initial or final control settings required prior to turning the equipment on or off.

3.3.4 Warm-Up Time - There shall be no required warm-up time before operating the equipment.

3.3.5 Vertical and Directional Gyro Signals - The equipment shall supply vertical and directional signals equivalent to the signals supplied by vertical and directional gyroscopes of the AN/AJB-3 Bombing System.

3.3.6 Air Data Computer Set Signals - The equipment shall supply signals equivalent to the signals supplied by the A/A24G-8 Air Data Computer Set and provide the gain scheduling capabilities of the A/A24G-8 Air Data Computer Set for the F4H-1 Aircraft.

3.3.7 Rate Gyro Signals - The equipment shall supply rate gyroscope signals of 0 to 12.6 ±1.5 volts, either In or Out of phase.

3.3.8 Aileron Transducer Signals - The equipment shall supply aileron transducer signals of 0 to 12.6 ±1.5 volts, either In or Out of phase.

3.3.9 Force Transducer Signals - The equipment shall supply force transducer signals of 0 to 12.6 ±1.5 volts, either In or Out of phase.

3.3.10 Lateral Accelerometer Signals - The equipment shall supply lateral accelerometer signals of 0 to 12.6 ±1.5 volts, either In or Out of phase.

3.3.11 Voltage Outputs - The Test Bench Harness shall supply voltages of 115, 66.5, 26, 15 and 8.7 volts, 400 cps, AC, 3 phase and 28 volts DC.

3.3.12 Lights - All lights used on the equipment shall be checked by a single switch when an AN/ASA-32 is connected to the Test Bench Harness.

3.3.13 Measurements - The equipment shall measure 0 to 20 ma differential current, 0 to 150 volts, 400 cps, AC, and 0 to 28 volts DC.

3.3.14 Test Points - Tip jacks shall be provided for all Autopilot and Aileron-Rudder Interconnect signal and switching circuits.

3.4 Detail Requirements

3.4.1 Test Set, Automatic Pilot Flight Controller TS-1592(*)/ASM-50 - The Test Set shall meet the following requirements:

3.4.1.1 Function - This assembly shall contain all the necessary controls for checking the Aileron-Rudder Interconnect System and the necessary signal simulators for checking the AN/ASA-32 when used with the Automatic Pilot Test Set TS-1593(*)/ASM-50.

3.4.1.2 Form Factor - The test set shall be contained in an aluminum dust cover which conforms to Specification MIL-T-21200. The maximum dimensions of the test set and dust cover shall not exceed 23.5 inches wide, 20 inches high and 9 inches deep.

3.4.1.3 Weight - The weight shall be kept to a minimum consistent with good design procedures and shall not exceed 75 pounds.

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3.4.1.4 Contents - The test set shall contain the following circuits and signal sources:

(1) Attitude Simulators - The Attitude Simulators shall supply signals equivalent to those received from Pitch Attitude, Roll Attitude and the Directional Gyro System. The above mentioned signals shall vary from 0 to 11.8 V AC, 400 cps, in phase and out of phase.

(2) Rate Simulators - The Rate Simulators shall supply signals simular to those received from the Pitch, Roll and Yaw Rate Gyro's in the AN/ASA-32. The above mentioned signals shall vary from 0 to 12.5 V AC, 400 cps, in phase and out of phase.

(3) Servo Test - The Servo Test shall provide signals for nulling the inputs to the Servo Amplifiers (Pitch, Roll and Yaw) under test. The above mentioned signals shall vary from 0 to 3.2 V, AC, 400 cps, in phase and out of phase.

(4) Aileron-Rudder Interconnect Simulator - The Aileron-Rudder Interconnect Simulator shall provide signals equivalent to that provided by the Aileron Transducers and simulates the switching accomplished by the Aileron-Rudder Interconnect (A.R.I.) System. The above mentioned signals shall vary from 0 to 12.5 V, AC, 400 cps, in phase and out of phase.

(5) Engage Limits and Interlocks - Simulated limiting switches shall be provided to assure that engaging interlock functions operate in sequence.

(6) Sensor Signal Sources - Simulated signals shall be provided equivalent to those received from the Lateral Accelerometer and the Force Transducer. The above mentioned signals shall vary from 0 to 12.5 V, AC, 400 cps, in phase and out of phase.

3.4.1.5 Controls - The test set shall contain the following controls which shall be located on the front panel:

(1) Pitch Gyro Test - The Pitch Gyro Test shall include a dial assembly labeled PITCH GYRO for supplying a simulated gyro signal and a two position switch labeled NORMAL-SIM PITCH GYRO for a selecting either the simulated gyro from the test set or an actual gyro under test.

(2) Roll Gyro Test - The Roll Gyro Test shall include a dial assembly labeled ROLL GYRO for supplying a simulated gyro signal and a two position switch labeled NORMAL-SIM ROLL GYRO for selecting either the simulated gyro from the test set or an actual gyro under test.

(3) Directional Gyro Test - The Directional Gyro Test shall include a dial assembly labeled DIR GYRO for supplying a simulated gyro signal and a two position switch labeled NORMAL-SIM DIR GYRO for selecting either the simulated gyro from the test set or an actual gyro under test.

(4) Lateral Accelerometer Test - The Lateral Acceleromater Test shall include a section labeled LAT ACCEL. In this section there shall be a three position switch with NORMAL-OFF-SIM markings and a knob labeled RUDDER for supplying simulated signals when the three position switch is in the SIM position. With the switch in the NORMAL position the signals from the AN/ASA-32 Lateral Accelerometer are checked. With the switch in the OFF position, all Lateral Accelerometer signals are disconnected from the test set.

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(5) Pitch Followup Sync. Dr. Test - The Pitch Followup Sync Dr Test shall include a signal source labeled PITCH FOLLOWUP SYNC DR and a two position switch labeled ON-OFF for checking the operation of the Pitch Followup Synchronizing Drive when the switch is in the ON position.

(6) Rate Gyro Test - The Rate Gyro Test shall include a section labeled RATE GYROS. Contained in this section shall be three knobs labeled PITCH, ROLL and YAW. In conjunction with the PITCH, ROLL and YAW labeling, three three position switches shall be included labeled NORMAL-OFF-SIM. With switches in OFF, all rate gyro signals are disconnected from test set; with switches in NORMAL the AN/ASA-32. Rate Gyro's are tested and in SIM simulated rate gyro signals are supplied by the test set.

(7) Servo Test - The Servo section shall be labeled SERVO TEST and shall contain three signal knobs labeled N. D. -N. U., WING DOWN and RUDDER to provide input null signals for checking the Servo Amplifiers in the AN/ASA-32.

(8) Aileron-Rudder Interconnect Test - The Aileron-Rudder Interconnect Test shall include a section labeled AIL. -RUD INTERCONNECT TEST. Contained in this section shall be three two position switches with the positions identified as NORMAL & SIMULATED and the switches being labeled as ARI CONTROL BOX, AIL. TRANSDUCER and SERVO LOAD. Included shall be one two position switch labeled AIRSPEED ABOVE 220 KT - BELOW 220 KT. There shall be two signal source knobs identified as AIL. TRANSDUCER LH - AIL. TRANSDUCER RH and Test Jacks for monitoring all functions of the A. R. I. System. The two AIL. TRANSDUCER knobs supply simulated signals for checking the A. R. I. 'S. when the AIL. TRANSDUCER switch is in the SIMULATED position.

(9) Accelerometer Test - The Accelerometer Test shall include a section labeled ACCELEROMETER TEST. This section contains a three position selector switch with the switch being labeled as ACCELEROMETER SELECTOR and the positions as G-LIMIT-OFF-AUTO TRIM with the switch function being to select either the G-LIMIT or AUTO TRIM accelerometer to check there limit switches. The section shall contain two lights labeled +4G and -1G for monitoring limit switch operation and a switch labeled RESET-OFF for reseting of the circuit used in checking the limit switches.

(10) Force Transducer Test - The Force Transducer Test shall contain a section labeled FORCE TRANSDUCER. Included in this section shall be a simulated transducer signal source knob with signal labeling of N. U. -N. D. and a three position switch labeled NORMAL-OFF-SIM for selecting the simulated signal or the transducer under test. It shall also contain three two position switches labeled RFS, PFS and EMER DISC which simulate these same switches in the Force Transducer.

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(11) Canceller Attenuator Test - The Canceller Attenuator Test shall contain a section labeled CANC-ATTN TEST. This section shall contain three switches labeled PITCH, ROLL and YAW with an OFF and TEST position for applying a know voltage to AN/ASA-32. Canceller and Attenuator amplifiers for gain checks.

(12) Engaging Controller Test - The Engaging Controller Test shall contain a section labeled ENGAGING CONTROLLER. This section shall contain four solenoid switches labeled STAB, AUG, AFCS, ATTITUDE and MACH equivalent to the solenoid switches in the AN/ASA-32 Engaging Controller.

(13) Limit and Interlock Switches Test - The Limit and Interlock Switches Test shall include a section labeled LIMIT & INTERLOCK SWITCHES. Included shall be the following limit and interlock switches: -1G, +4G, 0 to +3G, N.D., N.U., VGIR, ±90° PITCH and ±90 ROLL. These switches are used to check the AN/ASA-32 interlocks for operation.

(14) Servo Current Test - The Servo Current Test shall include a switch labeled SERVO CURRENT which is used with two current meters labeled EXTEND and RETRACT. The SERVO CURRENT switch shall have OFF, PITCH, PORT ROLL, STBD ROLL, YAW and ARI positions for checking servo amplifier output currents of the Pitch, Port Roll, Starboard Roll and Yaw of the AN/ASA-32 and the ARI of the A.R.I. system.

(15) A.C. Voltage Test - The A.C. Voltage Test shall include a meter labeled A.C. VOLTAGE and a switch labeled A.C. VOLTAGE for measuring the 3 phase A.C. Voltage input to the Test Set.

(16) Phase Test - The Phase Test shall include two lights for indicating phase rotation labeled PHASE-ABC and PHASE-ACB.

(17) D.C. Voltage Test - The D.C. Voltage Test shall include a meter labeled D.C. VOLTAGE and a section labeled POWER which includes a power light and switch to control the power supplied to the Test Set.

3.4.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
1J1	MS3110P-14-19S	Force Transducer Input
1J2	MS3110P-16-26S	Vertical Gyro Input
1J3	MS3110P-18-32S	Engaging Controller Input
1J4	MS3110P-20-41S	Aileron-Rudder Interconnect Input

3.4.2 Test Set, Automatic Pilot TS-1593(*)/ASM-50 - The Test Set shall meet the following requirements:

3.4.2.1 Function - This test set shall contain all the necessary jacks for checking the AN/ASA-32 Flight Control Group, the necessary simulation of the A/A24G-8 Air Data Computer Set for AN/ASA-32 operation. Servo actuator load simulation and AN/ASA-32 power simulation.

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3.4.2.2 Form Factor - The test set shall be contained in an aluminum dust cover which conforms to Specification MIL-T-21200. The maximum dimensions of the test set and dust cover shall not exceed 23 inches wide, 19.5 inches high and 9 inches deep.

3.4.2.3 Weight - The weight shall be kept to a minimum consistant with good design procedures and shall not exceed 75 pounds.

3.4.2.4 Contents - The test set shall contain the following circuits and signal sources:

(1) Air Data Computer Set Simulator - The Air Data Computer Set Simulator shall provide gain changing functions and signals equivalent to those provided by the Air Data Computer Set. The gain changing functions shall be as follows:

(a) Pitch Rate Vs Qc

(b) Stick Force Vs TAS

(c) Yaw Rate Vs MACH

(d) Pitch Attitude Vs Qc

(e) Lateral Acceleration

(f) Attitude Sensor

The simulated signals shall be Mach and Attitude sensors, and shall vary from 0 to 5 V AC, 400 cps, in phase and out of phase.

(2) Loads Simulator - The Loads Simulator shall provide simulated loads for the output of the AN/ASA-32 Servo Amplifiers (200 ohms for T Valve), Rate Gyro's (10,000 ohms), Force Transducer (10,000 ohms) and the Lateral Accelerometer (10,000 ohms).

(3) Power Source - The Power Source shall supply output voltages equivalent to those received from the AN/ASA-32 for checking AN/ASA-32 Components as follows:

	SIGNALS	A. C. or D. C.		JACKS	
1	Pitch Rate Gyro pickoff excitation	AC		903-9	903-10
2	Output of Pitch Rate Gyro pickoff	AC		903-18	903-28
3	Input to C. A. D. C. scheduling pot (pitch Rate Vs Qc)	AC		902-21	902-10
4	Output of C. A. D. C. scheduling pot (pitch Rate Vs Qc)	AC		902-32	Gnd
5	AN/AJB-3 (pitch pickoff) excitation	AC		905-1	905-8
6	Output of AN/AJB-3 (pitch pickoff)	AC	903-60	903-61	903-59
7	Input to CADC scheduling pot (pitch attitude Vs Qc)	AC		902-8	902-9
8	Output of CADC scheduling pot (pitch attitude Vs Qc)	AC		902-7	Gnd

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	SIGNALS	A. C. or D. C.		JACKS	
9	Output of Pitch Sync drive Amplifier	AC		906-43	906-42
10	Stick Force Sensor excitation	AC		901-34	901-27
11	Output of Stick Force Sensor	AC		901-57	901-38
12	Input to CADC Scheduling pot (stick Force Vs TAS)	AC		902-53	902-30
13	Output of CADC scheduling pot (stick Force Vs TAS)	AC		902-54	Gnd.
14	Output of pitch canceller	AC		906-11	Gnd.
15	Pitch actuator Aux. Ram Follow-up excitation	AC		904-39	904-40
16	Output of Pitch actuator Aux. Ram Follow-up	AC		904-62	904-63
17	Output of Pitch Servo Amplifier	DC	904-70	904-71	904-72
18	Pitch actuator Main Ram Follow-up excitation	AC		904-42	904-41
19	Output of Pitch actuator Main Ram Follow-up	AC		904-52	904-51
20	Pitch Bump input	AC		906-27	Gnd.
21	Output of Pitch Follow-up canceller	AC		906-26	906-25
22	Output of CADC attitude Sensor	AC		902-1	902-19
23	Output of CADC Mach No. Sensor	AC		902-3	902-40
24	Output of AN/AJB-3 Vertical Gyro Roll pickoff	AC	903-77	903-78	903-69
25	Output of Roll Sync drive Amplifier			905-17	905-16
26	Output of AN/AJB-3 Direction Gyro Pickoff	AC	901-2	901-1	901-3
27	Output of Course Sync drive Amplifier	AC		905-33	905-32
28	Port Roll actuator Follow-up excitation	AC		904-59	904-58
29	Output of Port Roll actuator Follow-up	AC		904-67	904-66
30	Output of Port Roll Servo Amplifier	AC	904-76	904-77	904-78
31	Output of Starboard Roll Servo Amplifier	AC	904-6	904-7	904-8
32	Starboard Roll Actuator Follow-up excitation	AC		904-13	904-14
33	Output of Starboard Roll Actuator Follow-up	AC		904-17	904-16
34	Roll Rate Gyro pickoff excitation	AC		902-58	902-57
35	Output of Roll Rate Gyro pickoff	AC		902-60	902-61
36	Yaw Rate Gyro pickoff excitation	AC		903-62	903-63

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	SIGNALS	A. C. or D. C.		JACKS	
37	Output of Yaw Rate Gyro pickoff	AC		903-51	903-52
38	Input to CADC scheduling pot (Yaw Rate Vs Mach No.)	AC		902-64	902-43
39	Output of CADC scheduling pot (Yaw Rate Vs Mach No.)	AC		902-73	Gnd.
40	Output of Yaw Series Servo Amplifier	DC	904-1	904-2	904-3
41	Yaw Series Actuator Follow-up excitation	AC		904-11	904-10
42	Output of Yaw Series Actuator Follow-up	AC		904-18	904-9
43	Lateral Accelerometer excitation	AC		903-44	903-32
44	Output of Lateral Accelerometer	AC		903-31	903-21
45	Input to CADC scheduling pot (Lateral accelerometer Vs Qc)	AC		902-33	902-13
46	Output of CADC scheduling pot (Lateral accelerometer Vs Qc)	AC		920-74	Gnd.
47	To Emergency disconnect switch	DC		901-68	Gnd.
48	From Emergency disconnect switch	DC		901-78	Gnd.
49	To Engaging controller Damper switch	DC		905-6	Gnd.
50	From Engaging controller Damper switch	DC		901-41	Gnd.
51	To Engaging controller Autopilot Engage switch	DC		901-42	Gnd.
52	From Engaging controller Autopilot Engage switch	DC		901-43	Gnd.
53	To Yaw series shut off valve	DC		904-20	
54	To pitch series shut off valve	DC		904-43	
55	To pitch parallel shut off valve	DC		904-32	
56	To Port Roll shut off valve	DC		904-46	
57	To starboard Roll shut off valve	DC		904-24	
58	To Engaging controller Mach.swtich	DC		901-63	
59	From Engaging Controller Mach switch	DC		901-62	
60	To Engaging Controller Attitude switch	DC		901-55	
61	From Engaging controller Attitude switch	DC		901-64	
62	To CADC Attitude Engage clutch	DC		902-18	902-70
63	To CADC Mach Engage clutch	DC		902-28	902-70

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	SIGNALS	A. C. or D. C.		JACKS	
64	From AN/AJB-3 Pitch Gimbal position switch	DC		903-45	Gnd.
65	From AN/AJB-3 Roll Gimbal position switch	DC		903-46	Gnd.
66	From AN/AJB-3 Directional Gyro slewing switch	DC		901-33	Gnd.
67	To Roll Rate Gyro Motor	AC	902-77	902-78	902-69
68	To Yaw Rate Gyro Motor	AC	903-70	903-71	903-72
69	To Pitch Rate Gyro Motor	AC	903-1	903-2	903-3
70	From CADC Static Pressure Compensator	DC		902-50	Gnd.
71	To G- Limit Accelerometer heater	AC		903-25	Gnd.
72	To Trim cut out accelerometer heater	AC		903-56	Gnd.
73	To G - Limit accelerometer switch (-1G & +5G)	DC		903-5	Gnd.
74	From G - Limit accelerometer -1G switch	DC		903-13	Gnd.
75	From G - Limit accelerometer +5G switch	DC		903-14	Gnd.
76	To Trim Cut out accelerometer switch	DC		903-75	Gnd.
77	From Trim Cut out accelerometer switch	DC		903-74	Gnd.
78	Phase A, 115 V AC, 400 cps	AC		901-5	Gnd.
79	Phase B, 115 V AC, 400 cps	AC		901-6	Gnd.
80	Phase C, 115 V AC, 400 cps	AC		901-7	Gnd.
81	Ground			901-17	Gnd.

(4) Input-Output Signal Check - The Input-Output Signal Check shall provide test jacks for checking all input and output signals, voltages and grounds of the AN/ASA-32.

3.4.2.5 Controls - The test set shall contain the following controls which shall be located on the front panel:

(1) Lamp Test - The Lamp Test shall include a switch labeled LAMP TEST for checking all lamps when a AN/ASA-32 Control Amplifier is connected to the test set.

(2) Simulated Amplifier Test - The Simulated Amplifier Test shall include a section labeled SIMULATED CONT AMPL ASSY. This section shall include an indicating light and switch labeled POWER for controlling the Simulated Cont. Amp. Assy. power. It shall also contain a switch labeled RATE GYRO for supplying simulated loads and power to the rate gyro's. Also a three position switch labeled FORCE TRANSDUCER-OFF-LATERAL ACCELEROMETER for supplying power and simulated loads to the Force Transducer and Lateral Accelerometer.

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(3) Air Data Computer Simulator Test - The Air Data Computer Simulator Test shall include a section labeled CADC. This section shall include pots for gain changing functions labeled PITCH RATE Vs Qc, STICK FORCE Vs TAS, YAW RATE Vs MACH, PITCH ATTITUDE Vs Qc, LATERAL ACCELEROMETER Vs Qc and ATTITUDE SENSOR GAIN. It shall also include two signal source knobs labeled MACH NO. SENSOR and ATTITUDE SENSOR for supplying simulated signals when the switch labeled SENSOR POWER is in the ON position. Also included is a switch labeled CADC INTLK SIMULATOR for supplying interlock voltage to the AN/ASA-32. Included are two monitoring light labeled ATTITUDE ENGAGE and MACH ENGAGE for checking when voltage is supplied to these circuits when checking the AN/ASA-32.

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(4) Servo Amplifier Loads Test - The Servo Amplifier Loads Test shall include a section labeled SERVO AMPLIFIER LOAD. Included in this section are four two position switches labeled PITCH, STBD. ROLL, PORT ROLL and YAW with the positions labeled SIMULATED and NORMAL. These switches are used for selecting either simulated loads supplied by the test set or actuator loads when connected to the test set.

(5) Switching Circuitry Test - The Switching Circuitry Test shall include monitoring lights labeled EDS, -1G, +4G, CADCIR, DER, VGIR, ER, RLS, IR, PFS, RFS, OG +3G, AER and MER for checking operation of the AN/ASA-32 switching and interlock circuitry.

(2) Power, Signal and Ground Tests - The Power, Signal and Ground Tests shall include test jacks for monitoring all power, signal and grounds which are brought out to external connectors on the AN/ASA-32. The Test Set labeling shall be J901, J902, J903, J904, J904 and J906 corresponding to the connectors on the AN/ASA-32. Under each of the J901, J902, J903, J904, J905 and J906 labeling sections shall be the jacks numbered corresponding to the connector pin numbers.

3.4.2.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
2J5	MS3110P-20-41S	Simulated or Actual CADC Input
2J6	MS3110P-22-55S	Actuator's Input
2J7	MS3110P-20-41SW	Rate Gyro's Input
2J420	MS3110P-20-41S	Simulated CADC Output

3.4.3 Rack, Amplifier MT-2595(*)/ASM-50 - The amplifier rack shall meet the following requirements:

3.4.3.1 Function - The amplifier rack shall provide a mounting for the AN/ASA-32 Control Amplifier and connect the Control Amplifier to the Test Harness Circuitry.

3.4.3.2 Form Factor - The amplifier rack shall be of aluminum construction conforming to Specification MIL-T-21200. The maximum dimensions shall not exceed 13 inches wide, 6.5 inches high and 21 inches long.

3.4.3.3 Weight - The weight of the amplifier rack shall not exceed 15 pounds.

3.4.4 Bracket, Limiting Accelerometer MT-2596(*)/ASM-50 - The Bracket shall meet the following requirements:

3.4.4.1 Function - The Limiting Accelerometer Bracket shall be used for mounting the Limiting Accelerometer when testing the operation of the limit switches.

3.4.4.2 Form Factor - The Limiting Accelerometer Bracket shall be of aluminum construction conforming to Specification MIL-T-21200 and the maximum dimensions shall not exceed 9.5 inches long, 5.5 inches wide and 6 inches high.

3.4.4.3 Weight - The weight of the Limiting Accelerometer Bracket shall not exceed 5 pounds.

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3.4.5 Bracket, Lateral Accelerometer MT-2597(*)/ASM-50 - The Lateral Accelerometer Bracket shall meet the following requirements:

3.4.5.1 Function - The Lateral Accelerometer Bracket shall be used for mounting the Lateral Accelerometer when testing for signals from this unit.

3.4.5.2 Form Factor - The Lateral Accelerometer Bracket shall be of aluminum construction conforming to Specification MIL-T-21200. The maximum dimensions shall not exceed 6.5 inches long, 6 inches wide and 5 inches high.

3.4.5.3 Weight - The weight of the Lateral Accelerometer Bracket shall not exceed 5 pounds.

3.4.6 Mount, Rate Gyroscope MT-2598(*)/ASM-50 - The Rate Gyroscope Mount shall meet the following requirements:

3.4.6.1 Function - The Rate Gyroscope Mount shall be used to install the Pitch, Roll and Yaw Rate Gyro's when testing for signal output.

3.4.6.2 Form Factor - The Rate Gyroscope Mount shall be of aluminum construction conforming to Specification MIL-T-21200 and the maximum dimensions shall not exceed 9.0 inches long, 4.5 inches wide and 4.5 inches high.

3.4.6.3 Weight - The weight of the Rate Gyroscope mount shall not exceed 10 pounds.

3.4.7 Cable Assembly, Special Purpose, Electrical W-1 of AN/ASM-50 - W-1 of AN/ASM-50 shall meet the following requirements:

3.4.7.1 Function - W-1 of AN/ASM-50 shall be used to connect the Test Harness to a Central Air Data Computer (CADC) adapter when required.

3.4.7.2 Form Factor - The cable assembly shall be constructed of wire conforming to Specification MIL-W-16878 and shall be 90 inches long.

3.4.7.3 Weight - The weight of the cable assembly shall not exceed 4 pounds.

3.4.8 Cable Assembly, Special Purpose, Electrical W-2 of AN/ASM-50 - W-2 of AN/ASM-50 shall meet the following requirements:

3.4.8.1 Function - W-2 of AN/ASM-50 shall be used for connecting the Test Harness to the Electrohydraulic actuators when testing these units.

3.4.8.2 Form Factor - The cable assembly shall be constructed of wire conforming to Specification MIL-W-16878 and shall be 120 inches long.

3.4.8.3 Weight - The weight of the cable assembly shall not exceed 6 pounds.

3.4.9 Cable Assembly, Special Purpose, Electrical W-3 of AN/ASM-50 - W-3 of AN/ASM-50 shall meet the following requirements:

3.4.9.1 Function - W-3 of AN/ASM-50 shall be used for connecting the AN/ASA-32 Rate Gyro's to the Test Harness when testing these units.

3.4.9.2 Form Factor - The cable assembly shall be constructed of wire conforming to Specification MIL-W-16878 and shall be 72 inches long.

3.4.9.3 Weight - The weight of the cable assembly shall not exceed 3 pounds.

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3.4.10 Cable Assembly, Special Purpose, Electrical W-4 of AN/ASM-50 - W-4 of AN/ASM-50 shall meet the following requirements:

3.4.10.1 Function - W-4 of AN/ASM-50 shall be used for connecting the Simulated CADC section of the Test Harness to the CADC input when running tests on the AN/ASA-32.

3.4.10.2 Form Factor - The cable assembly shall be constructed of wire conforming to Specification MIL-W-16878 and shall be 18 inches long.

3.4.10.3 Weight - The weight of the cable assembly shall not exceed 2 pounds.

3.4.11 Cable Assembly, Special Purpose, Electrical W-5 of AN/ASM-50 - W-5 of AN/ASM-50 shall meet the following requirements:

3.4.11.1 Function - W-5 of AN/ASM-50 shall be used for connecting the AN/AJB-3 Bombing System Vertical and Directional Gyro signal outputs to the Test Harness.

3.4.11.2 Form Factor - The cable assembly shall be constructed of wire conforming to Specification MIL-E-16878 and shall be 72 inches long.

3.4.11.3 Weight - The weight of the cable assembly shall not exceed 4 pounds.

3.4.12 Cable Assembly, Special Purpose, Electrical W-6 of AN/ASM-50 - W-6 of AN/ASM-50 shall meet the following requirements:

3.4.12.1 Function - W-6 of AN/ASM-50 shall be used for connecting the Engaging Controller to the Test Harness when testing this component.

3.4.12.2 Form Factor - The cable assembly shall be constructed of wire conforming to Specification MIL-W-16878 and shall be 24 inches long.

3.4.12.3 Weight - The weight of the cable assembly shall not exceed 2 pounds.

3.4.13 Cable Assembly, Special Purpose, Electrical W-7 of AN/ASM-50 - W-7 of AN/ASM-50 shall meet the following requirements:

3.4.13.1 Function - W-7 of AN/ASM-50 shall be used for connecting the A.R.I. System Control Box and Transducers when testing these components.

3.4.13.2 Form Factor - The cable assembly shall be constructed of wire conforming to Specification MIL-W-16878 and shall be 72 inches long.

3.4.13.3 Weight - The weight of the cable assembly shall not exceed 4 pounds.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, including the Sand and Dust Test and modifying the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with Specification MIL-I-6181. Drop tests for transit and combination cases shall be in accordance with Specification MIL-T-21200.

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4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests -When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Reliability Quality Assurance Tests

(3) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.4.2 Reliability Quality Assurance Tests - Equipments which have successfully passed the individual tests shall be tested for reliability quality assurance in accordance with Specification MIL-R-23094.

4.4.2.1 Test Level - The Test level shall be Level II (from Table I of Specification MIL-R-23094).

4.4.2.2 ACCEPT-Reject Criteria - Figure 12 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria.

4.4.2.3 Detailed Reliability Test Procedure - The contractor shall prepare the detailed reliability test procedure and submit it to the procuring agency for approval. The procedure shall include, but not be limited to, the following:

(a) Accept-Reject Criteria

(b) Details of the test cycle.

(c) Performance characteristics to be measured

(d) Failure Criteria

(e) Preventive Maintenance to be allowed

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

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4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Use - The equipment described in this specification is intended to be used for bench tests, calibration and maintenance of the AN/ASA-32 Flight Control Group and the F4H-1 Aircraft Aileron-Rudder Interconnect System. It shall simulate any components necessary to test any of this equipment.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±.5V, D.C. and 115 ±1V, A.C. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.5 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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